SAMPLE AUDIT | 113,425 ORDERS | BRAZILIAN E-COMMERCE MARKETPLACE

Olist: late deliveries hurt reviews, and almost nobody orders twice

Where one marketplace’s growth stalled, what the data shows about why, and what to test first.

The short version

Olist’s revenue grew fast through 2017 and peaked at nearly R$1 million in November. Then it stalled. From March to August 2018 it stayed between R$838K and R$977K a month, growing 1% to 2%.

Two problems sit underneath. Late deliveries pull review scores down hard. And 96.9% of customers never place a second order, so revenue depends on a constant supply of new customers.

The data does not show that late delivery is why customers do not come back. Customers with good deliveries and high reviews left too, and every customer segment averages about 4 stars or better. These are two separate problems, and this report treats them that way.

The numbers

Customers who never placed a second order, out of 96,076 Observed in the data	**96.9%**
Average review score, on-time orders against late orders Observed in the data	**4.2 vs 2.5**
Customers acquired in November 2017 who bought again in December Observed in the data	**40 of 7,060**
Average review score range across all five customer segments Observed in the data	**4.0 to 4.3**

Observed figures are measured directly in the dataset. **Hypothesis** marks an explanation or action the data cannot confirm.

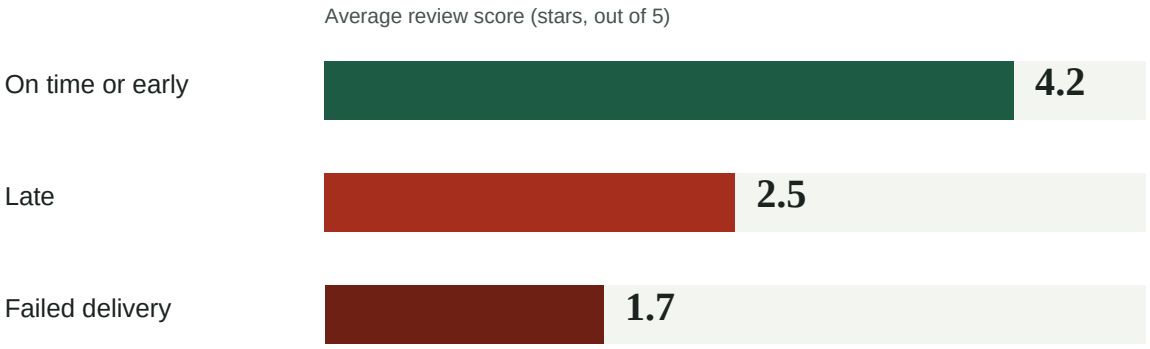
What to do, in order

- **Fix delivery where it fails:** remote northern states, bulky items, and the highest-earning sellers with the worst late rates.
- **Test a second-purchase offer** on the Promising segment, against a control group that gets nothing.
- **Match the message to why customers left.** Customers whose order arrived late need an apology and a fix. Satisfied one-timers need a different follow-up.

> **BEFORE ACTING**
> Check whether customers whose first order arrived late reorder less than customers whose first order arrived on time. That comparison is what would link the two problems, and this report does not claim it. Details on pages 3 and 4.

1. Late delivery cuts review scores nearly in half

Orders that arrive on time or early average 4.2 stars. Late orders average 2.5. Orders that fail to arrive average 1.7.



Average review score by delivery outcome. Observed in the data.

Where delivery goes wrong

- **Geography.** Delays concentrate in the north of Brazil. Amazonas and Roraima are among the worst states.
- **Product type.** Large and bulky items, such as office furniture, take the longest to arrive.
- **Sellers.** Some of the ten highest-earning sellers also have some of the worst late-delivery rates.

Shipping cost hurts reviews too. In some categories, Electronics for example, freight is sometimes about 70% of the item's price. Those orders drew low reviews even when they arrived on time.

WHAT THIS MEANS

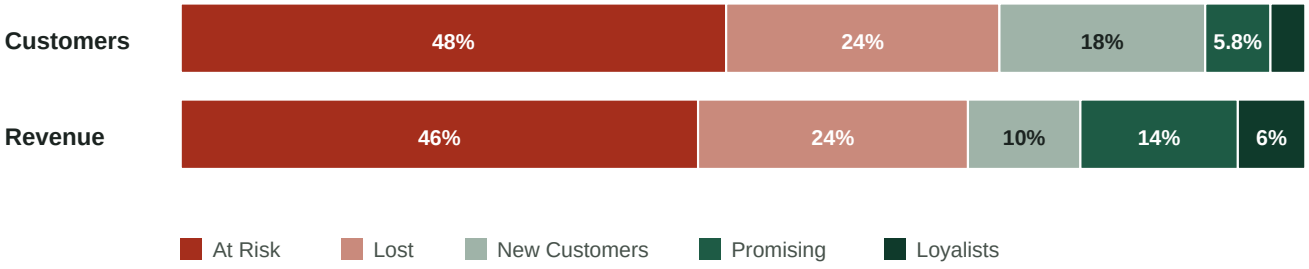
Poor delivery is a review-score problem you can locate by state, product type and seller. That makes it fixable in pieces rather than as one vague logistics project. It does not yet tell you whether fixing it brings customers back. Page 3 covers why.

2. Almost nobody orders twice

96.9% never ordered again **3.1% ordered more than once**

Of 96,076 unique customers, 2,994 (3.1%) ordered more than once. Observed in the data.

The November 2017 Black Friday peak shows it. Olist acquired 7,060 new customers that month. 40 of them bought again in December. By the third month, 12 had.



Share of customers and share of revenue by segment. Revenue shares are rounded from the segment revenue totals. At Risk and Lost bring in about 70% of revenue because they are 72.7% of customers. Promising is 5.8% of customers and about 14% of revenue. Loyalists are 3.1% of customers and about 6% of revenue.

Who the customers are

Segment	Customers	Spend per customer	Average review	Days ahead of estimate	Who they are
At Risk	46,553 (48.5%)	R$160	4.0	9.5	One-time buyers who last ordered a while ago.
Lost	23,321 (24.3%)	R$165	4.1	12.7	One-time buyers who last ordered long ago.
New Customers	17,608 (18.3%)	R$90	4.3	11.1	One-time buyers who ordered very recently, with ordinary spend.
Promising	5,595 (5.8%)	R$405	4.1	13.2	One-time buyers who ordered very recently and spent a lot.
Loyalists	2,994 (3.1%)	R$315	4.1	11.6	Customers who ordered more than once.

Spend, review and delivery figures are segment averages. **Days ahead of estimate** is the gap between the promised and the actual delivery date, so higher is better. Segments are scored on recency, frequency and spend with custom thresholds, because with about 97% of customers buying once, standard quintile buckets would be meaningless.

- **Promising customers are worth the most per head:** about R$405 from a single order, against about R$315 in total for Loyalists across all their orders.

WHAT THE DATA DOES AND DOES NOT SHOW

Supported: late delivery lowers review scores, from 4.2 stars on time to 2.5 when late.

Weak: a link between delivery and the At Risk segment. At Risk ranks last on reviews and delivery margin, but by small gaps, and its orders still arrived early on average.

Not supported: that late delivery is why customers do not return. Every segment averages about 4 stars or better, including Lost customers, who had good deliveries. Non-return is spread across satisfied customers, which points away from satisfaction as the main cause.

The check that would settle it: compare the second-order rate of customers whose first order arrived on time with those whose first order arrived late. If late first deliveries reorder far less, delivery is part of the retention problem. If not, retention needs its own fix.

What to do

1. Fix delivery where it fails

ACTION	Set realistic delivery estimates for the remote northern states and for bulky items. Put the highest-earning sellers with the worst late rates on a delivery improvement target. Find which carriers serve the worst states and review them.
WHY	Observed: the review gap, and the concentration of delays by state, product type and seller.
HOW TO KNOW IT WORKED	Share of orders delivered after the estimate, and average review score, in the affected states and for the affected sellers.
NOTE	Longer estimates make fewer orders count as late without making them faster, so track actual delivery days alongside. This dataset has no carrier field, so the carrier review needs Olist’s own logistics records.

2. Test a second-purchase offer on Promising customers

ACTION	Give Promising customers a reason to reorder soon after a successful first delivery, for example 15% off valid for 30 days. Run it against a control group that gets no offer, and against a second discount size. Repeat the test on New Customers, a group about three times larger.
WHY	Hypothesis. Promising customers are worth the most per head, about R$405 from one order, and have not come back. The dataset has no price-response data, so the right discount is unknown and has to be tested.
HOW TO KNOW IT WORKED	Share placing a second order inside the window, against the control, and margin after the discount.

3. Match the message to why customers left

ACTION	Split At Risk by what happened to their order. **Order arrived late:** acknowledge the delay and offer a fast-shipping voucher, not a plain discount, and only where delivery has actually improved. **Order arrived on time:** a standard win-back message. **Lost:** skip discounts and try catalogue-awareness campaigns for higher-frequency categories.
WHY	Hypothesis. At Risk orders arrived about 9.5 days ahead of the estimate on average, so an apology for delay would be wrong for most of them. The segment average hides who was actually let down.
HOW TO KNOW IT WORKED	Reorder rate by group, against an untreated sample.

Lower priority: bundles for Loyalists. They are 3.1% of customers and about 6% of revenue, and spend less per customer than Promising, so even a good result moves little. The dataset also has no margin data, so “high-margin” bundles cannot be chosen from it.

Limits of this analysis

- Public Kaggle data, not client data. Olist is a marketplace with many third-party sellers, so a single Shopify brand will differ.
- Observational. Everything here is an association, not a proven cause. No experiments were run.
- A customer with one order counts as not returning. The dataset covers a limited window, so some later returns are missed.
- Segment figures are averages. A segment can average 4 stars and still contain unhappy customers.
- No cost, margin or marketing-spend data. The report cannot say what customers cost to acquire or whether an offer pays for itself.

How this was done. Python for cleaning and exploration across five relational tables, MySQL for five business queries (CTEs and window functions), and a four-page Power BI dashboard. Code and queries are public: github.com/zeeshan-akram-ds/Olist-ECommerce-Analysis

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